Fiscal Year:	FY 2008	**Task Last Updated:**	FY 02/22/2008
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Dynamical Selection of Three-Dimensional Interfacial Patterns in Directional Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	08/01/2006	**End Date:**	03/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Shan Liu, who worked on the project for several years has left Iowa State University and will not be involved in the project. Dr. Myung-Jin Suk is working partly on this project and carrying out experimental studies.		
COI Name (Institution):	Karma, A (Northeastern University)		
Grant/Contract No.:	NNM04AA47G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The proposed research will consist of accurate and rigorous numerical modeling of the dynamics of pattern evolution, coupled with critical ground-based and low gravity experimental studies in a model transparent system in which the dynamics of pattern evolution can be observed in situ. Initial ground-based studies in the succinonitrile-salol and succinonitriole-water systems have been extended to succinonitrile-camphor system that will be used in DECLIC experiments to establish the range of conditions over which cellular structures are stable under diffusive growth conditions. To achieve the objectives of the proposed research, accurate and rigorous numerical modeling of the dynamics of pattern evolution is developed along with critical ground-based experimental studies in model transparent system in which the dynamics of pattern evolution can be observed in situ. Ground-based experiments have been carried out in the succinonitrile-camphor system in thin samples to establish the range of conditions over which cellular and		

	dendrtic structures are stable under diffusive growth conditions. Since Succinonitrile-camphor dydtem will be used in DECLIC, phase diagram and physical constants of this systems have been accurately determined.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of fundamental quantitative knowledge of microstructure evolution under dynamic growth conditions is crucially important for improving large scale industrial processes from casting to welding. The results will provide basis to predict reliable processing conditions that optimize microstructures and hence material properties and performance. In addition, the knowledge gained from this study will be applied to other pattern forming phenomena that occur in different fields of science such as combustion and fluid physics. The benchmark data obtained under low gravity conditions will be a cornerstone for the validation of phase-field approach through unique quantitative comparison of model predictions and benchmark experiments in three dimensions.
Task Progress:	The following important aspects have been examined in this grant period. • We have developed an experimental technique to study pattern formation in capillary samples in which diffusive growth conditions are present. In thin samples we have obtained a single cell in the succinonitrile-camphor system. This system has a eutectic that forms behind the cells so that precise shape and tip temperature of the cell can be accurately determined. We have shown a growth regime in which the single cell can be described by the model developed by the three-dimensional extension of the Saffmann and Taylor model for viscous fingering. We have shown that the shape of the cell matches accurately with the model, thereby showing the similarity between cellular growth and finger growth in different disciplines of sciences. We have characterized the transition regime in which the experimental results show a transition from the finger to a parabolic cell shape. • The experiments on the single cell also allowed us to measure the cell tip temperature accurately over a range velocity, thermal gradient and composition. We found that the tip temperature results deviate significantly from the predictions of the current models that are widely used in the literature. We have developed a Saffmann-Taylor model to obtain an expression for the tip temperature that agrees closely with our experiments. This work is being dome in collaboration with Dr. Bernard Billia, who is the principal investigator of the DECLIC experiments that will be carried out by CNES on International Space Station. • Experimental results also show self-similarity of cellular shapes in an array, which gives scaling between the spacing and the tip radius of the cells. The tip temperature and self-similar cell shapes have also been investigated by using the rigorous phase-field simulations that predict the experimental results obtained in the succinonitrile-camphor system. • Since succinonitrile-camphor system will be used in DECLIC experiments, we have accurately characterized the phase diagram of this system, and determined the value of the diffusion coefficient. The accurate values of these parameters are critical in comparing the experimental results with the theoretical model. • Since most ground-based experiments in transparent system are carried out in thin samples to obtain diffusive growth conditions, we have completed a detailed study of the effect of thickness on microstructure evolution. Significant changes in microstructures have been found when the thickness becomes very small. The characteristics of cells and dendrites were found to change significantly and the conditions for transition in microstructures were also found to depend on the thickness. • Detailed study of the effect of sample thickness on eutectic structure in succinonitrile-camphor was carried out. It was found the when the thickness becomes smaller, the shape of the rod eutectic phase changes from circular to elliptical. Two different orientations of the ellipse were observed: with major axis in the direction of the thickness and in the direction of the width of the sample. The orientation of the ellipse was found to change when velocity or thickness we changed. A detailed analytical model of elliptic cylindrical rods was developed that predicted the observed results. In addition, we showed that the elliptic rod eutectic is a general eutectic growth model which converges to circular rod eutectic when the major and the minor axis of the ellipse are equal and to a lamellar eutectic when one of the axes is much larger than the other one. • A significant progress has been made in the phase-field modeling of cellular microstructures. We have developed the codes that can predict the dynamic and steady-state cellular interface pattern evolution. The phase-field model is now being extended to an array of cells in three-dimensions. It is found that the anisotropy in interface energy, which is extremely small in the succinonitrile-camphor system (less than one percent), plays a crucial role in the pattern evolution. For isotropic interface energy no steady-state is reached. A very small anisotropy value quickly stabilizes the pattern and gives a steady-state growth. The characteristics of cells, such as tip radius and cell spacing significantly depend on the value of the anisotropy in the system. • We have carried out detailed experiments in the SCN-camphor system to establish the conditions that would be optimum for DECLIC experiments. This work was done in collaboration with Dr. Billia who is the PI of the CNES part of the project.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Liu S, Li J, Lee JH, Trivedi R. "Spatio-temporal microstructure evolution in directional solidification processes." Philosophical Magazine, 2006;86(24):3717-38. http://dx.doi.org/10.1080/09500830500504036 , Nov-2006
Articles in Peer-reviewed Journals	Athreya P, Dantzig J, Liu S, Trivedi R. "On the role of confinement on solidification in pure materials and binary alloys." Philosophical Magazine, 2006;86(24):3739-56. http://dx.doi.org/10.1080/14786430500157060 , Nov-2006
Articles in Peer-reviewed Journals	Teng J, Liu S, Trivedi R. "Growth and morphology of rod eutectics." Acta Materialia 2008 Jul;56(12):2819-33. http://dx.doi.org/10.1016/j.actamat.2008.02.011 , Jul-2008
Articles in Peer-reviewed Journals	Teng J, Billia B, Trivedi R. "Cellular microstructure in directional solidification." Physical Review E (submitted), February 2008. , Feb-2008